

MA10021:Mathematics for Engineers

UNIT 1

Contributions of Indian mathematicians

1. Baudhayana

- ❖ Baudhayana was an Indian Mathematician who was born in 800 BC and died in 740 BC.
- ❖ He was a Vedic brahmin priest.
- ❖ He is said to be the original founder of Pythagoras's Theorem.
- ❖ Baudhāyana discovered Pythagoras at least 1000 years before Pythagoras was born.
- ❖ **Baudhayana** composed the Sulbasutra to provide guidelines for religious ceremonies.
- ❖ The Sulbasutras is like a guide to the Vedas which formulate rules for constructing altars. In other words, they provide techniques to solve mathematical problems effortlessly.
- ❖ **Baudhayana theorem**
A shloka from the Śulbasûtra is proof that he had the concept of [Pythagoras theorem](#) in his mind even before the Pythagoras was actually made:

दीर्घस्याक्षणया रज्जुः पार्श्वमानी, तिर्यग्दं मानी, च यत्पृथग्भूते
कुरुतस्तदुभयाङ्करोति।

It means “A rope stretched along the length of the diagonal produces an area which the vertical and horizontal sides make together.”

- ❖ He was amongst the first-ever Indian Mathematicians who calculated the value of pi.

- ❖ According to Baudhayana, the approximate value of π was 3. In his Sulba Sutra, he used different values of π for different geometric constructions, especially when drawing circular shapes.
- ❖ Baudhayana also provided an approximate value of the square root of 2, one of the earliest known attempts at solving irrational numbers. Baudhayana describes the method of finding the length of a diagonal given the lengths of the sides of a square. In other words, it describes the method of finding the square root of 2. To get the value of the diagonal of a square, by adding one-third to the side, then adding one-fourth of it, then subtracting thirty-fourth of it, That is it's diagonal approximately. That is 1.4142161, which is correct to five decimals.
- ❖ Bodhayana even tried to find out
 - a circle whose area is the same as that of a square.
 - Diagonals of a rhombus bisect at right angles,
 - Diagonals of a rectangle bisect each other,
 - The midpoints of a rectangle joined forms a rhombus whose area is half the rectangle, etc.
 - The area of a square formed by joining the middle points of a square is half of the original.

2. Aryabhata

- ❖ Aryabhata was an ancient Indian mathematician and astronomer who lived during the Gupta dynasty, approximately between 476 and 550 CE.
- ❖ Aryabhata was one of the brightest minds of ancient India, a legendary mathematician and astronomer whose pioneering ideas shaped the world's understanding of numbers, planetary motion, and the very nature of time.
- ❖ Aryabhata, also known as the father of Indian Mathematics, was a renowned astronomer and mathematician of the ancient times of India.

❖ Aryabhata contributed significant work in science and mathematics and concluded theories of earth rotation on its axis, approximation of pie (π) place value system of zero, trigonometry, and many others.

Aryabhattha's Mathematical Contributions:

- **Decimal System & Place Value:**

Aryabhata was instrumental in developing the place value system and using zero as a placeholder, which were foundational to the decimal system.

- **Approximation of Pi:**

He calculated an approximation for Pi as 3.1416 (using the formula: $(8 \times 100 + 62000) / 20000$), Add four to 100, multiply by eight, then add 62,000. Divide by 20,000." This gives $\pi \approx 3.1416$, which was remarkably accurate for his time and is considered a major contribution.

- **Trigonometry:**

He introduced concepts of sine and cosine and created tables of sines, a major advancement in the field.

- **Algebra:**

Aryabhata's work included methods for solving quadratic equations and formulating algebraic identities, earning him the title "Father of Algebra".

- **Geometric Formulas:**

He correctly calculated the areas of triangles and circles, and formulated formulas for the volume of spheres and pyramids.

- **Arithmetic Series:**

He formulated simplified formulas for summing series of squares and cubes, as well as for finding square and cubic roots.

Astronomy Contributions of Aryabhat:

Principle of Rotation: The discovery, recorded in the Aryabhatiya, that the Earth rotates around its own axis from west to east is significant.

- *Aryabhatta also declared that the Earth rotates around the sun and the moon moves round the earth.*
- *Eclipses: In Aryabhatiya he introduces the idea of shadows, cast by and falling on earth, moon, and planets, and states that the lunar eclipse is caused by the entering of the moon into the earth's shadow.*
- *Aryabhatta gives formulas for the length and diameter of the earth's shadow, the timing and duration of the eclipses, and or the size of the eclipsed part of the sun or moon.*
- *Circumference of the Earth: Aryabhatta also revealed that the circumference of the Earth is 39,968km....*
- *It is 40,072 km according to modern scientific calculations....*
- **Eclipses:** He provided scientific explanations for the occurrence of solar and lunar eclipses.

Length of the Year: He calculated the length of a year with exceptional precision.

He also described the number of days in a year to be 365.

3. Varahamihira

- ❖ Astrologer, astronomer, and polymath Varahamihira, born in c. 505 and passed away in c. 587, also known as Varaha or Mihira.

- ❖ He wrote several books on these topics, including the well-known "Pancha Siddhantika," which deals with the five systems of astronomy.
- ❖ He also wrote "Brihat Samhita," which covers astrology, architecture, planetary motions, cloud formation, eclipses, timekeeping, rainfall, seasons, agriculture, and many other subjects.

Varahamihira's Contributions to Mathematics

- ❖ Varahamihira is widely recognised for his contributions to trigonometry (trikonamiti) formulae.
- ❖ He invented new methods for calculating the sine (jya) and cosine (kojya) functions, as well as the cotangent and versine functions.
- ❖ Further, He improved the accuracy of Aryabhata's sine tables.
- He developed the algebraic properties of zero and negative numbers.
- He was also one of the first mathematicians to discover a different form of Pascal's triangle.
- It was used to compute the binomial coefficients.
- His book Brihat-Samhita presented a systematic treatment of algebraic equations, including quadratic equations.
- He was the first to discover the 4x4 magic square.

Other Contributions

- **Varahamihira** also made several significant discoveries in the fields of hydrology, geology, and ecology.
- His claim that termites and plants can sense subsurface water is increasingly receiving support among scientists.
- Additionally, he wrote a lot. His command of **Sanskrit language** and poetic rhythm allowed him to express himself in a distinctive manner.

- He became well-known due to his **encyclopaedic knowledge** and engaging presentation of even the most boring subjects, including astronomy.
- He has been promoted to the same level of astrology as **Kautilya, Manu, and Panini** in political philosophy, law, and grammar thanks to his works like the **Pancha Siddhantika** (Five Principles), **Brihat samhita** (Master Collection), and **Brahjataka** (Astrological study).

Medieval Era: Brahmagupta , Bhaskara II, Madhava of Sangamagrama

4. Brahmagupta

- ❖ Brahmagupta (597- 668 AD) was ,one such genius Astronomer – Mathematician ,born in a city called Bhinmal in Rajasthan.
- ❖ Brahmagupta is also known as Bhillamal Acharya.
- ❖ His contributions to mathematics are found in two works, the first of which *Brahmasphutasiddhanta* (BSS) and second was titled *Khandakhayaka*.

Brahmagupta Contributions to Mathematics

1. Zero and Negative Numbers:

His most famous and well-known contribution to the mathematics world is his definition and use of zero with negative numbers.

2. Algebra:

- Brahmagupta provided solutions for general linear equations and quadratic equations.
- He worked on indeterminate equations, including Pell's equation ($x^2 - ny^2 = 1$), and found integer solutions for equations of the form $ax + by = c$.

3. Geometry:

- He made significant contributions to the study of cyclic quadrilaterals (four-sided figures inscribed in a circle).
- Brahmagupta's formula for calculating the area of a cyclic quadrilateral is still recognized today.
- He also explored properties of diagonals and perpendiculars in cyclic quadrilaterals.

4. Trigonometry:

- Brahmagupta developed formulas for calculating trigonometric functions like the sine and cosine.
- He also gave an approximate value for pi (π).

5. Other Notable Contributions:

- He calculated the length of a year as 365 days, 6 hours, 12 minutes, and 19 seconds, which is very close to the modern value.
- Brahmagupta also discussed the concept of gravity, stating that bodies fall towards the earth because of its attractive force.

5. Bhāskara II

- ❖ **Bhāskara II** (C. 1114–1185), also known as **Bhāskarāchārya** ('Bhāskara the teacher'), was an Indian polymath, mathematician, and astronomer.
- ❖ His works, particularly *siddhanta Siromani* and *Lilvati*, continue to be appreciated for their depth and clarity.
- ❖ Bhaskara's arithmetic text ("Lilavti") covers the topics of definitions, arithmetical terms, interest computation, arithmetical and geometrical

progressions, plane geometry, solid geometry, the shadow of the gnomon, methods to solve indeterminate equations, and combinations.

- ❖ His Bījaganita ("Algebra") was a work in twelve chapters. It was the first text to recognize that a positive number has two square roots (a positive and negative square root).
- ❖ The Siddhānta Shiromani (written in 1150) demonstrates Bhaskara's knowledge of trigonometry, including the sine table and relationships between different trigonometric functions.

Some of Bhaskara's contributions to mathematics include the following:

- A proof of the Pythagorean theorem by calculating the same area in two different ways and then cancelling out terms to get $a^2 + b^2 = c^2$.
- In Lilavati, solutions of quadratic, cubic and quartic indeterminate equations are explained.
- Solutions of indeterminate quadratic equations (of the type $ax^2 + b = y^2$).
- Integer solutions of linear and quadratic indeterminate equations (Kuṭṭaka). The rules he gives are (in effect) the same as those given by the Renaissance European mathematicians of the 17th century.
- A cyclic Chakravala method for solving indeterminate equations of the form $ax^2 + bx + c = y$. The solution to this equation was traditionally attributed to William Brouncker in 1657, though his method was more difficult than the chakravala method.
- The first general method for finding the solutions of the problem $x^2 - ny^2 = 1$ (so-called "Pell's equation") was given by Bhaskara II.
- Solutions of Diophantine equations of the second order, such as $61x^2 + 1 = y^2$. This very equation was posed as a problem in 1657 by the French mathematician Pierre de Fermat, but its solution was unknown in Europe until the time of Euler in the 18th century.

- Solved quadratic equations with more than one unknown, and found negative and irrational solutions.
- Preliminary concept of mathematical analysis.
- Preliminary concept of differential calculus, along with preliminary ideas towards integration.
- preliminary ideas of differential calculus and differential coefficient.
- Stated Rolle's theorem, a special case of one of the most important theorems in analysis, the mean value theorem. Traces of the general mean value theorem are also found in his works.
- Calculated the derivative of sine function, although he did not develop the notion of a derivative. (See Calculus section below.)

6. Mādhava of Sangamagrāma (Mādhavan)

- ❖ **Mādhava of Sangamagrāma (Mādhavan)** (c. 1340 – c. 1425) was an Indian mathematician and astronomer who is considered to be the founder of the Kerala school of astronomy and mathematics in the Late Middle Ages.
- ❖ Madhava made pioneering contributions to the study of infinite series, calculus, trigonometry, geometry and algebra.
- ❖ He was the first to use infinite series approximations for a range of trigonometric functions.
- ❖ Among his many contributions, he discovered many methods for calculating the circumference of a circle. One of Madhava's series is known from the text *Yuktibhāṣā*, which contains the derivation and proof of the power series for inverse tangent, discovered by Madhava.
- ❖ Madhava composed an accurate table of sines. Madhava's values are accurate to the seventh decimal place. Marking a quarter circle at twenty-four equal intervals, he gave the lengths of the half-chord (sines) corresponding to each of them.

- ❖ Madhava's work on the value of the mathematical constant Pi is cited in the *Mahajyānayanaprakāra* ("Methods for the great sines").

Evolution of Indian Mathematics and Its Contributors:

Modern era: Ramanujan, Shanti SwaroopBhatnagar, Harish Chandra

7. SrinivasaRamanujan (1887-1920)

- ❖ **SrinivasaRamanujan** (born December 22, 1887, Erode, India—died April 26, 1920, Kumbakonam) was an Indian mathematician known for his contributions to infinite series, number theory and game theory .and the properties of the partition function.
- ❖ His geniusness was that he discovered his own theorems. He proved over 3,000 theorems, identities and equations, including on highly composite numbers, partition function and its asymptotes, and mock theta functions.
- ❖ Due to his great achievements in the field of Mathematics, Indian govt. decided to celebrated his birthday 22nd December as Mathematics Day.

Ramanujan's Contribution

- **Hardy-Ramanujan Number**

Once Hardy and Ramanujan arrived at Ramanujan's residence in a taxi cab having number 1729. Hardy quoted that the number 1729 seemed to be uninteresting but Ramanujan promptly replied that this was a very interesting number as it is the smallest number which can be expressed as the sum of cubes of two numbers in two different ways as given below :

1729 it is the smallest number which can be expressed as the sum of cubes of two numbers in two different ways as given below:

$$1729 = 1^3 + 12^3 = 10^3 + 9^3$$

Later some theorems were established in theory of elliptic curves which involves this fascinating number.

- **Infinite Series For π :**

Srinivasa Ramanujan also discovered some remarkable infinite series of π around 1910. The series

$$\frac{1}{\pi} = \frac{2\sqrt{2}}{9801} \sum_{k=0}^{\infty} \frac{(4k!)(1103 + 26390k)}{(k!)^4 396^{4k}}$$

computes a further eight decimal places of π with each term in the series. Later on, a number of efficient algorithms have been developed by number theorists using the infinite series of π given by Ramanujan.

- **Goldbach's Conjecture**

Goldbach's conjecture is one of the important illustrations of Ramanujan contribution towards the proof of the conjecture.

The statement is every even integer > 2 is the sum of two primes, that is, $6=3+3$.

Ramanujan and his associates had shown that every large integer could be written as the sum of at most four primes (Example: $43=2+5+17+19$).

- **Theory of Equations**

Ramanujan was shown how to solve cubic equations in 1902 and he went on to find his own method to solve the quadratic. He derived the formula to solve bi-quadratic equations.

He also worked in some other areas of mathematics such as congruence method known as Ramanujan's Congruences, highly composite numbers, hypergeometric series, Bernoulli numbers, Fermat's last theorem. He focused mainly on developing the relationship between partial sums and products of hypergeometric series. He independently discovered Bernoulli numbers and using these numbers, he formulated the value of Euler's constant up to 15 decimal places. He nearly verified Fermat's last theorem which states that no three natural number x, y and z satisfy the equation $x^n + y^n = z^n$ for any integer $n > 2$.

8. Shanti Swaroop Bhatnagar

- ❖ Shanti Swarup Bhatnagar was a prominent Indian scientist, born on February 21, 1894, and died on January 1, 1955.
- ❖ Shanti Swarup Bhatnagar is renowned for his contributions to science and technology, particularly in chemistry and the establishment of India's scientific infrastructure, rather than directly in mathematics.
- ❖ While he collaborated with physicists on the Bhatnagar-Mathur Magnetic Interference Balance and played a key role in building post-independence science and technology in India, his primary focus was on chemistry and its applications.
- ❖ Shanti Swarup Bhatnagar's first industrial problem was developing the process for converting bagasse (peelings of sugarcane) into food-cake for cattle.

Shanti Swarup Bhatnagar contributions:

- **Magnetochemistry and Emulsions:** Bhatnagar made significant contributions to magnetochemistry and the physical chemistry of emulsions.

- **Bhatnagar-Mathur Magnetic Interference Balance:** He co-developed this highly sensitive instrument for measuring magnetism.
- **National Research Development Corporation (NRDC):** He played a crucial role in establishing the NRDC.
- **Advocacy for Industrial Research:** He championed the cause of industrial research and played a role in establishing oil refineries in India.
- **Founding Director of CSIR:** He was the founder director of the Council of Scientific & Industrial Research (CSIR).

While his work didn't directly focus on mathematics, his overall contributions to science and technology laid a strong foundation for scientific development in India, which certainly benefited the mathematical community as well. The Shanti Swaroop Bhatnagar Prize for Science and Technology is awarded in his memory to recognize outstanding contributions in various scientific disciplines, including mathematical sciences.

9. Harishchandra

- ❖ **Harish-Chandra** (11 October 1923 – 16 October 1983) was an American mathematician and physicist who did fundamental work in representation theory, especially harmonic analysis on semi simple Lie groups.
- ❖ After a short while, Harish-Chandra went to Cambridge, where he studied for his Ph.D. under Dirac's supervision. During his time in Cambridge, he moved away from physics, and became more interested in mathematics.
- ❖ Dirac visited Princeton for one year, and Harish-Chandra worked as his assistant during this time. However, he was greatly influenced by the mathematicians Hermann Weyl and Claude Chevalley. The period from 1950 to 1963 was his most productive, and he spent these years at the

Columbia University. During this time, he worked on representations of semi simple Lie groups.

Awards

He was a member of the [National Academy of Sciences](#) and a [Fellow of the Royal Society](#). He was the recipient of the [Cole Prize](#) of the [American Mathematical Society](#), in 1954. Harish-Chandra worked at the Institute for Advanced Study at Princeton from 1963. He was appointed IBM von Neumann Professor in 1968. The [Indian National Science Academy](#) honoured him with the [Srinivasa Ramanujan Medal](#) in 1974.

The Indian Government named the [Harish-Chandra Research Institute](#), an institute dedicated to Theoretical Physics and Mathematics, after him.